

Project Name: SmartERP Agentic Framework

Topic: How AI Agents are Transforming ERP Systems: From Claude MCP to Open Code and Autonomous Agents

Transforming ERP Systems with AI

1. Traditional ERP Systems

Traditional ERPs rely on manual navigation, complex menus, and fixed buttons to retrieve data.

2. Proprietary Context Protocols (Claude MCP)

Closed, third-party protocols connect external AI tools well, but they often incur perquery subscription costs, vendor lockin, and data privacy risks by sending sensitive business data to external servers.

3. Custom Open-Code Architecture

Moving to "Open Code" means leveraging custom software integration and open-ecosystem AI models instead of paid proprietary frameworks.

General Benefits: Keeps sensitive enterprise data 100% private on internal enterprise servers while eliminating recurring API subscription fees.

4. Autonomous Background Agents

AI agents that evolve beyond reactive, prompt-based chatbots by autonomously monitoring background business workflows (e.g., stock levels) and triggering necessary operational processes without waiting for human prompts.

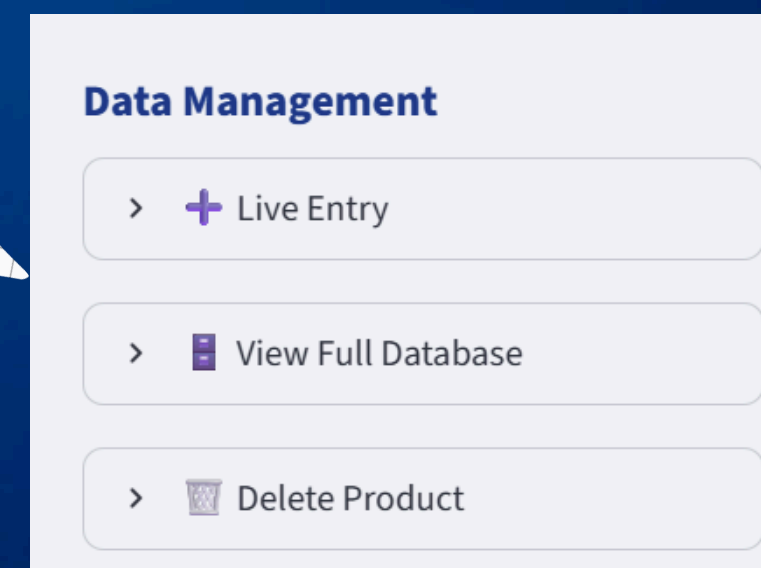
How the SmartERP Web App Works

What Is This App?

An intelligent ERP dashboard built to replace traditional, click-heavy software with direct, natural language interaction.

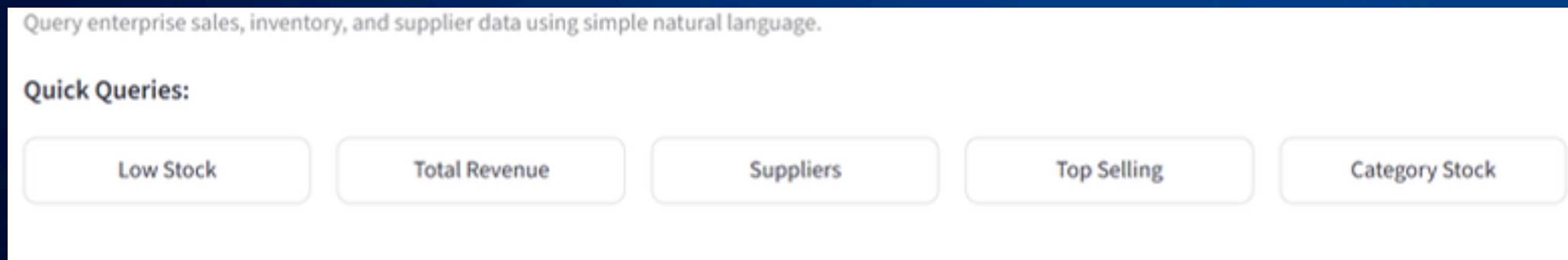
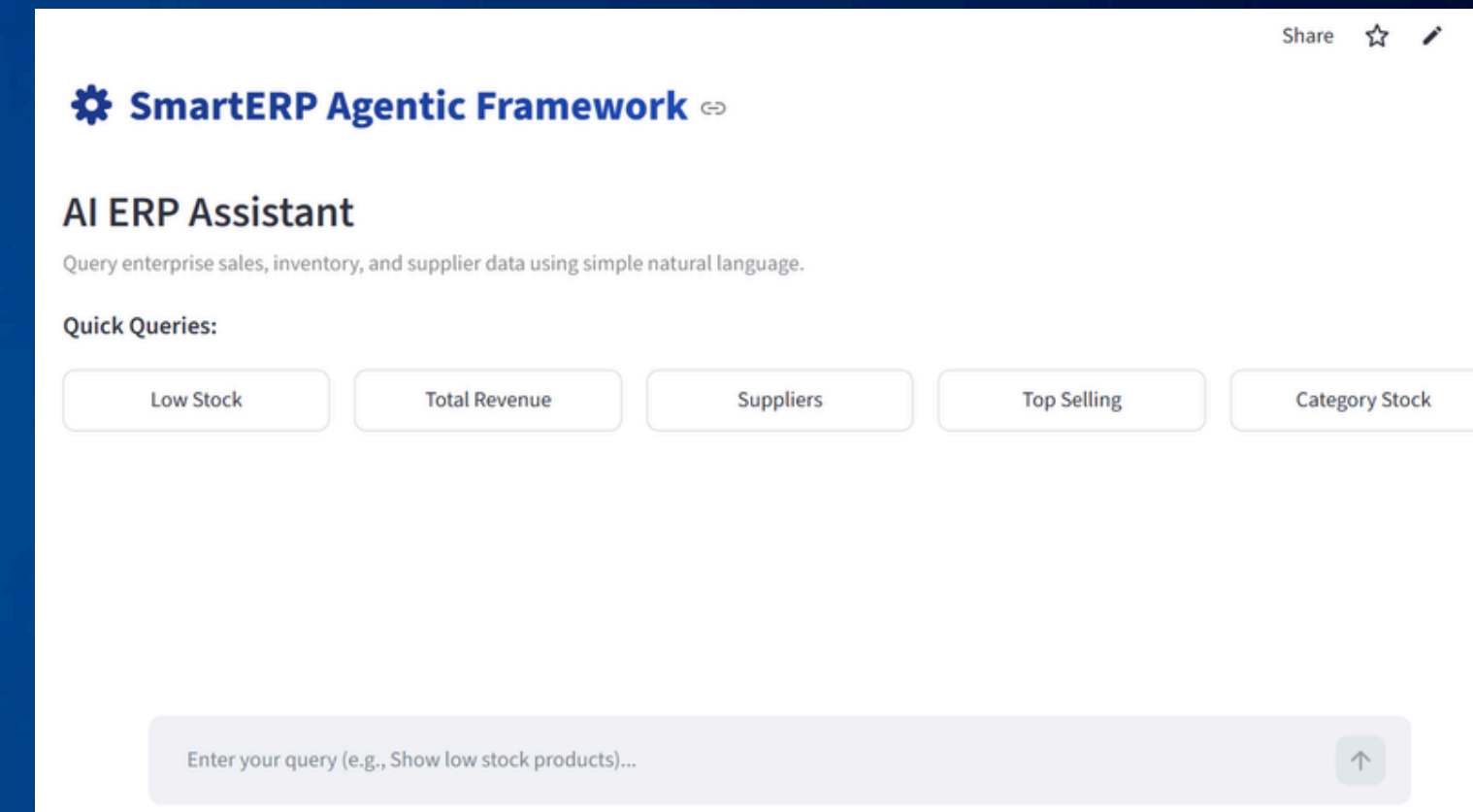
Executive Analytics & Live Data Management

- ***Instant KPI Metrics:*** Displays real-time metrics for Total Products, Stock Valuation, Sales Revenue, and Low-Stock Alerts.
- ***Interactive Visualizations:*** Dynamic Plotly charts update automatically as sales and inventory change.
- ***Live Data Operations (Why DB Features Exist):*** Includes manual Data Entry, Live Table View, and Product Deletion so users can test live data changes and watch the dashboard refresh instantly.



2. AI ERP Assistant (Natural Text-to-SQL)

- **No-Code Querying:** Translates plain English questions into executable SQL without writing code.
- **Domain-Restricted Queries:** Safely restricted to internal database tables (inventory, sales, suppliers) and will not attempt to answer unrelated general knowledge.
- **Built-in Sample Shortcuts:** Features clickable sample prompt buttons for fast, error-free testing.



3. Autonomous Operations & Safety Thresholds

- ***What is Minimum Threshold?***

Every product has a preset Min Threshold representing its safety stock limit.

- ***Smart Agent Detection:*** The autonomous background agent automatically scans inventory, detects any product falling below its threshold, and instantly drafts supplier purchase orders.

- ***Human-in-the-Loop Control:*** Managers review requisitions and click Approve or Reject, updating the status visually while keeping database records safe.

SmartERP Agentic Framework

Autonomous Operations & Procurement Hub

Automated inventory monitoring and purchase order generation system.

Execute Stock Audit

Alert: 4 item(s) below safety stock threshold. Action required.

Pending Purchase Requisitions ⇄

Product	Current	Min Threshold	Recommended Action	Status / Action
Industrial Lubricant	120	200	Drafted to: ChemTech Supplies (supplier2@example.com)	<button>Approve</button> <button>Reject</button>
Stretch Wrap Film	80	150	Drafted to: Polymer World Bangladesh (info@polymerworld.bd)	<button>Approve</button> <button>Reject</button>

 Live Web Application :

<https://erpframework.streamlit.app/>

Github link:

<https://github.com/SanjidaSadiah30/SmartERP-Agentic-Framework>

Technology Stack Used :

- **User Interface:** Streamlit with Custom CSS (Sleek, responsive single-screen layout).
- **Database Engine:** SQLite3 (Lightweight, local enterprise storage).
- **AI Intelligence:** Llama 3.3 via Groq SDK (High-speed open-weight LLM).
- **Data Analytics:** Pandas & Plotly (Real-time charts and data processing).

Open Architecture & System Boundaries

The "Open Code" Advantage (vs. Proprietary Claude MCP):

- **No Third-Party Protocol:** Built with custom Python connectors instead of closed, vendor-locked protocols like Claude MCP.
- **Zero Subscription Cost:** Runs on open-weight AI infrastructure with no monthly dollar costs per query.
- **Complete Data Privacy:** Keeps enterprise inventory and financial records 100% inside private local servers.

System Boundaries & Safety Controls:

- **Strict Knowledge Limits:** The AI model is strictly bounded by the database schema; it does not answer out-of-domain queries.
- **Read-Only AI Agent:** The AI agent only generates SELECT queries to prevent accidental database corruption, leaving data modifications strictly to human-controlled UI forms.

The background is a solid dark blue. In the upper left, there is a faint, light blue wireframe of a globe. In the lower half, there are several horizontal, wavy bands of different shades of blue, creating a layered, ocean-like effect.

Thank You